Collin Atkins

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Education

Bachelor of Science in Aerospace Engineering, May 2027 — The University of Alabama - Tuscaloosa, AL
President's List (2025); Dean's List (2023, 2024, 2025) — GPA: 3.73/4.00

Relevant Projects

Independent Project Lead for NAR L1 Certification Rocket, ARA, Tuscaloosa, AL, Sep 2026- Present

- Designing and modeling an H100W_DM-4 configuration rocket in OpenRocket and SOLIDWORKS to evaluate vehicle geometry, flight performance, and stability.
- Implementing a parachute recovery system incorporating shock cords for a safe descent and vehicle recovery.

Stability and Control Lead for Senior Design Project, University of Alabama, Tuscaloosa, AL, Aug 2026 – Present

- Leading the stability and control development of a VTOL drone designed to operate in Titan's atmosphere, land on its hydrocarbon seas, and collect liquid samples.
- Developing parameterized MATLAB and Simulink models to analyze drone flight dynamics, stability, and attitude-control performance under Titan flight conditions.
- Conducting a PID/LQR controller trade study and developing risk mitigation strategies and verification plans.

Independent Project Lead for Aerospace Structures Analysis, University of Alabama, Tuscaloosa, AL, Apr 2026

- Developed a computational model of an unsymmetrical L-shaped cantilever beam using Bernoulli-Euler beam theory to calculate section properties, bending stresses, neutral-axis orientation, and deflections.
- Compared theoretical predictions with experimental strain and deflection measurement data to reduce differences by refining Young's modulus and clamping boundary conditions.

Aerodynamics Analyst/Project Co-Lead for NACA Airfoil Analysis, University of Alabama, Tuscaloosa, AL, Apr 2026

- Analyzed NACA airfoil geometries using Thin Airfoil Theory and XFOIL by performing inviscid and viscous simulations to evaluate lift characteristics, pitching moments, and pressure distributions across varying AOAs.
- Generated lift-curve, pressure-coefficient, and mean-camber-line plots in Excel to compare the aerodynamic performance of the NACA airfoils and document the findings in a technical report.

Team Lead for Glider Project, University of Alabama, Tuscaloosa, AL, Oct 2023 – Dec 2023

- Led an aerospace student engineering team in the design, construction, and testing of a functional glider and established design requirements, constraints, and performance objectives.
- Coordinated project scheduling, budgeting, and team responsibilities using a Gantt chart and WBS.

Experience

Shadow, NASA, Airbus, and International Paper, Huntsville, AL/Columbus, MS, May 2024 - Aug 2025

- Observed engineering operations at NASA (Huntsville, AL), Airbus (Columbus, MS), and International Paper (Columbus, MS), gaining direct exposure to professional aerospace and industrial engineering environments.

Dishwasher, Zacharies, Columbus, MS, May 2022 - Aug 2022

- Managed dish pit operations and supported waitstaff to maintain kitchen workflow.

Activities and Awards

- National Society of Leadership and Success (NSLS), Oct 2026 - Present
- The University of Alabama AIAA Student Chapter, Sep 2026 – Present
- The University of Alabama Rock Climbing Club, Sep 2026 – Present
- Alabama Rocketry Association (ARA) IREC Team, Aug 2026 – Present
- UA AI Experience, Aug 2026
- American Institute of Aeronautics and Astronautics (AIAA), Apr 2026 – Present
- Salutatorian, May 2023
- BSA Eagle Scout, Jun 2022
- Volunteer Leader for Park Cleanup and Renovation, Oct 2022
- 200+ hours of volunteer service, Aug 2019 - Present

Skills

Engineering Tools: XFOIL, MATLAB, SOLIDWORKS, ABAQUS, OpenRocket, ANSYS FLUENT, Python, Simulink
Software: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Microsoft Teams, Miro, GanttPRO, Google Workspace
Soft Skills: Project Management, Technical Documentation, Problem-Solving